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W3 Day 45

Memo No. _____

Date 1 / 1 / 11

API (Application Programming Interface) is a set of rules and protocols that allow different software applications to communicate and exchange data with each other.

API Test case Design

i) Positive Test Cases.

ii) Negative Test Cases

iii) Boundary Value and Edge Case Testing

iv) Data Driven Testing - Execute the same test case with multiple sets of input data to validate API behavior across different data combination.

v) Validation Check - Verify the response format, required field, data types, status code, headers and business rules.

Tunneling. A technique of sending data through an intermediate network or server to create a logical connection between two endpoints.

API Testing Process

- Understand API requirement

It will understand what the APIs supposed to do, what which endpoints are available.

- Identify Test Scenario
- Prepare the Test Environment
- Create Test Cases
- Send API Request
- Validate API Responses
- Execute API Tests
- Analyze and Report Defects
- Retest and Perform Regression Testing
- Generate Test Reports

Structure of a Test Case

Test ID

Test Name / Description

Endpoints

Preconditions

Request Headers

Request body

Expected status

Expected Result

Actual Result.

Some Important HTTP Status code

100 - Continue

101 - Switching Protocols

200 - Successful

201 - Created

204 - No Content

300 - Multiple Choices

302 - ~~Found~~ Found

305 - Use Proxy

306 - Unused (Reserved but not in use)

307 - Temp redirected

308 - Permanent Redirect

400 - Bad request

401 - Unauthorized

402 - Payment Required

403 - Forbidden

404 - Not Found

405 - Method not Allowed

406 - Not Acceptable

408 - Request Time out

429 - Too Many Requests

431 - Request Header Field Too Large

428 - Pre Condition Required

500 - Internal Server Error

502 - Bad Gateway

503 - Service Unavailable

504 - Gateway Timeout

511 - Network Authentication Required.

Stateless and Statefull

- Server doesnot maintain client session information between requests

- Server maintains client / session information between request -

Elements of Header

i) Authorization

It sends authentication credentials / token

eg. JWT or Basic Auth

ii) Content type

Tells the server what format the request body is in (JSON)

iii) Accept

It tells the server what response format

The client wants.

iv) User-Agent

Identifies the client making the request, such as browser, or mobile app.

v) Host-~~Agent~~

Specifies the server/domain receiving the request

vi) Cookie

Sends stored cookies, commonly used for session authentication

vii) Set Cookie

Server uses this to create/update a cookies in the client

viii) Origin

Shows the origin (scheme + host + port) from which a request originated, important for CORS

ix) Referer

Indicates the page URL that led to the current request

x) Accept-Encoding

Tells the server which compression formats the client supports such as gzip.